

gives a brilliant metallic surface. Hardness about 2 of Mohs' scale.

In the open tube, before the blowpipe flame, the mineral fuses quietly, coloring the glass a bright yellow under the assay; a white or gray sublimate is deposited at a short distance from, or directly over it, which, on being heated, fuses into transparent drops, resembling oil. On charcoal it fuses readily to a leaden-colored globule, which, on cooling, becomes covered with little points or dendrites. This globule flattens under the hammer, but breaks on the edges. With the addition of a little carbonate of soda, a globule of silver is readily obtained. A fragment heated to redness in a closed tube or in a trass, with dry carbonate of soda and charcoal dust gives, on the addition of a few drops of boiling water, the beautiful violet-red or purple solution described by Berzelius as characteristic of tellurium.—This solution loses its color after standing for some time, and a dark colored powder is deposited. The mineral dissolves in hot nitric acid, with the separation of tellurous acid in crystals.

It is probably the species *Hessite*, but the decision is reserved until further examinations are made. Its color is darker than the Hessite of Savodinsky, Siberia, and is not quite so hard.

This very rare mineral has not hitherto been observed in America, and its occurrence is therefore of peculiar interest. I am indebted to P. C. Currier, Esq., of Georgetown, for the specimen. It was obtained in that vicinity, and probably taken from the auriferous drift; but it cannot have been transported far from its original source.

The crystals give reactions for lead and sulphur and a trace of selenium. They are probably galena, but may contain tellurium.

A specimen seen in California in 1854, weighing about two ounces, greatly resembled the massive part of the specimen above described. The small fragment of it which was then obtained, also gives the reactions for tellurium and silver. Its precise locality is not known.

The telluric silver of Siberia, according to Gustaf Rese, is composed, in 100 parts, of:

|                 |       |
|-----------------|-------|
| Tellurium,..... | 36 96 |
| Silver,.....    | 62 42 |
| Iron,.....      | 0.24  |

It is probable that tellurium combined with silver, lead or bismuth, will be found in the auriferous quartz of Grass Valley and other localities. A few specimens in my possession contain small brilliant grains resembling tetradymite, but their exact character is not yet determined.

January 1, 1857.

A letter was read from M. Boissudal, stating that *Saturnia Californica* had been previously described by him as *S. euryalus*.

The curators were authorized to send such duplicates of specimens as are available, to the Society of Natural History, at Stockton, Cal.

SAN FRANCISCO, March 30, 1857.

President in the Chair.

Prof. Asa Gray, of Cambridge, Mass., and Dr. John Torrey, of New York, were elected Honorary Members.

Dr. John Browne, of the U. S. steamer *Active*, Mr. Geo. Gibbs, of Port Townsend, M. Moreenhaut, of Monterey, Capt. Fauntleroy, Capt. Wilson and Dr.

George Suckley, were elected Corresponding Members.

#### *Donations to the Cabinet.*

From Capt. J. D. Brown, of the schooner *Ada*, specimens of Reptiles, Fishes, Annelida and Crustacea, from the Gulf of California.

From Dr. Pigné Dupuytren, a fine collection of Marine Shells and a skull of *Belone*, from New Caledonia; also, a fabric made from the hair of the bat, by the natives of those islands.

From Dr. Welch, a skull of *Diomedea chlororhynchus*, and a skull of a native of the Sandwich Islands.

From Dr. Czapkay, a large species of *Patella*.

From Mr. Parent, two specimens of *Chiton* and a *Cularis*, from the Gallipagos Islands.

From Mr. H. P. Wakelee, a suite of specimens from Nicaragua, consisting of the capsule of Cocoa Bean, Scorpion, skin of a large serpent, (called by the natives Bo-bo,) and Coal; also, Copper smelted at the La Mina del Padre, (Colima,) and Copper Ore from La Mina Truxcanisco, (Colima); also, Solar Salt from near Los Angeles, Cal.; also Auriferous Quartz from Kern river, and a shell of *Echinus*.

From Mr. H. R. Bloomer, a specimen of *Eutainia dorsalis*, from San Francisco.

From Barry & Patten, a specimen of *Ornithorynchus*, from Australia.

The thanks of the Academy were voted for the donations above recorded.

From Dr. Lanzweert, Liguite, from Sonoma.

From Dr. Bennett, a specimen of Gypsum, and one of native Alum, from Guaymas.

From Capt. Russell, a fine specimen of *Yucca*, in full bloom, from the Southern Coast; also, two young specimens of the same for cultivation; also, several specimens of *Mesembryanthemum*.

From Dr. Kellogg, a specimen of *Scolopendra*, from Monte Diatolo; also, a box of seeds.

From Dr. Ayres, a specimen of *Anarrhichthys ocellatus*, and one of *Cebidichthys crista-galli*, from San Francisco Bay.

From Mr. W. H. Pease, of Honolulu, a fine suite of specimens from the Sandwich Islands, consisting of Crustacea, Land and Marine Shells, Echinoderms and Corals.

From Mr. Hepburn, Calcareous Travertine, from the Geysers.

From Mr. Beardslee, a specimen of *Trillium*.

From Dr. George Suckley, specimens of *Mytilus*, from the Straits of Fuca.

From Dr. Trask, a specimen of *Nereis*, thirty-three inches in length, from San Francisco Bay; also, a fish allied to *Gunnellus*, of apparently a new generic type; also, a quantity of the ripe fruit of the Coffee Tree, from the Sandwich Islands. The curators were requested to distribute these seeds throughout the State, for the purpose of inducing their cultivation.

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ON THE DIRECTION AND VELOCITY OF THE EARTHQUAKE IN CALIFORNIA, JANUARY 9, 1857—BY DR. JOHN B. FRANK.

The earthquake which occurred in various parts of this State, on the morning of the 9th January last, excited at the time considerable attention.—This arose from two causes. First, from the varied reports that appeared on the following day through the press of this city, detailing its occurrence in remote mountain towns, and for which there was no foundation. Secondly, from the great extent over which the commotion was felt, as was subsequently proved.

Immediately following the occurrence of the phenomenon, letters were addressed to all the principal towns between Mariposa and Downieville, east of the valleys, for the purpose of learning how far the shocks may have extended eastward of this city.—The letters were forwarded by the Pacific Express Company to their agents, and through them answers were returned in every case but two through the same source. From the facts thus obtained, it was found that in no locality east of the foothills, *was any shock felt whatever on that day or night*.

Another report, equally unfounded, reached us on the arrival of the steamer from the Southern coast, to the effect that several houses had been demolished in San Diego from its violence, while the facts in the case are, *that the steamer left that port twenty-four hours before the shock occurred there*.

This earthquake, or more properly speaking the series of shocks that began on the night of the 8th in this city, and which continued in the south part of the State during the following day and night of the 9th, was probably the most extensive of any on record on this portion of the Pacific coast, excepting, perhaps, that of the wave of the Simoda earthquake in December, 1854. The linear distance over which we are able to trace its course, amounts to six hundred and two miles, and its breadth, so far as now ascertained, is two hundred and ninety miles. It has all the appearance of having been the terminal movement of some more violent commotion at a distance from our coast.

From the best evidence obtainable at present, it seems to have had its origin to the west and travelled in an easterly direction. This is conclusively proved from the fact that it was felt earlier at San Francisco than at any other locality east of this city within the State. We have no record as yet of its occurrence along the coast of Mexico or Oregon.

I have been able to determine with considerable accuracy the period of time at which the shock between eight and nine o'clock on the morning of the 9th took place, at four localities east of the city of San Francisco, in this State, as the shock at that hour seems to have been more generally noticed than those which either preceded or followed it here or elsewhere, though at this city it was much less marked than the shocks at 1h. 33m., 4h. 15m., and 7h., these three latter occurring at those hours of the morning when most persons are sleeping. The shock at 7h. produced a circular motion in the pendulum, the diameter of which was about five inches. The oscillations of the pendulum in all the others were in an easterly and westerly direction.

The precise period of time at which the shock took

place at San Francisco, between eight and nine o'clock, is determined by the stopping of a time-piece belonging to J. W. Tucker, whose rate of error was three seconds fast. The time at San Diego was furnished by Mr. Cassidy, of the army, and that of the Tejon Reserve is by persons at that post. To private gentlemen at Sacramento and Stockton we are indebted for the time at those places. The accompanying table of latitudes and longitudes, of localities named gives the hour at which the shock took place at each; the difference or elapsed time, from which the velocity was deduced, are the mean times corrected for the places named, the time as given above being taken as the standard at San Francisco.

It is proper here to state that three minutes four seconds, was the greatest error in time found, and the least was twenty two seconds:—

|                   | Lat.  | Lon.   | Time of shock. | Elapsed time. | Velocity |
|-------------------|-------|--------|----------------|---------------|----------|
|                   | °     | '      | h. m. s.       | m. s.         | miles    |
| San Francisco.... | 37 43 | 122 25 | 8 13 30        | 0 00          | 0 0      |
| Sacramento.....   | 38 32 | 121 23 | 8 29 00        | 7 30          | 6 6      |
| Stockton.....     | 37 52 | 121 34 | 8 23 00        | 9 30          | 6 5      |
| Tejon.....        | 35 00 | 118 46 | 8 45 00        | 32 30         | 6 0      |
| San Diego.....    | 32 42 | 117 13 | 8 50 00        | 36 30         | 7 0      |

The velocity is given in miles per minute, and by dividing the sum of the same by their number, it will be found that the movement of the wave at that time averages a fraction over 6.2 miles per minute.

The results obtained from the above data approximate closely the deductions of Prof. Bache on the wave which reached our shores resulting from the earthquake at Simoda on the 23d December, 1854, and which will be found in a paper read by that gentleman at the meeting of the American Association for the Advancement of Science, during the early part of last year.

From the facts before us, there can be but little doubt of the direction of the commotion, and that it proceeded from the west, or a little south of that point. The motion of the earth, as described at the different localities at which it was felt, with the motion of the pendulum—which was slightly south of a west line—leads to the latter conclusion. Time is an important element in aiding us to form correct conclusions regarding their phenomena, and it is to be hoped that our friends in different parts of the State, in reporting the same, will be precise in this particular. Of the moments attending the shocks, many and varied reports have reached us; and it seems to have acted with greater violence in the vicinity of the Tejon Reserve and upper Tulare county than at any other places. It is most remarkable that so small an amount of intensity was manifested when the area over which it extended is taken into consideration.

The effects were felt in San Francisco several hours before they are reported to have been observed at any other place north or south. They began here at twenty minutes past eleven, on the night of the 8th, and continued till thirteen minutes past eight the following morning—six shocks occurring in the interim; while to the south, the first shock that was noticed at the Tejon was at 6 hours 30 minutes, on the 9th. In Los Angeles they continued at long intervals through the day until 23 hours 30 minutes of the same date. I have learned from persons who were present in Los Angeles at this time, and also at the shock of the 14th July, 1855, that the severity of the latter exceeded that of the 9th January last past.

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